9.1 Vector Functions

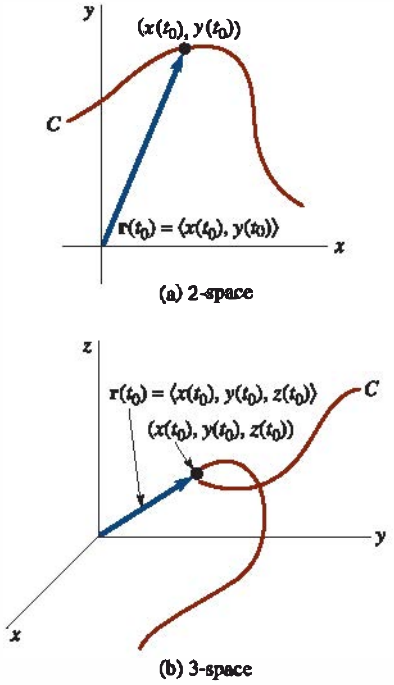

FIGURE 9.1.1 Curves defined by vector functions

Introduction

Recall that a **curve** C in the xy-plane is simply a set of ordered pairs (x, y). We say that C is a **parametric curve** if the x- and y-coordinates of a point on the curve are defined by a pair of functions $x = f(t)$, $y = g(t)$ that are continuous on some interval $a \le t \le b$. The notion of a parametric curve extends to 3-space as well. A **parametric curve in space**, or **space curve**, is a set of ordered triples (x, y, z) where

$$x = f(t), \quad y = g(t), \quad z = h(t), \tag{1}$$

are continuous on an interval defined by $a \le t \le b$. In this section we combine the concepts of parametric curves with vectors.

Vector-Valued Functions

It is often convenient in science and engineering to introduce a vector $\mathbf{r}$ whose components are functions of a parameter t. We say that

$$\mathbf{r}(t) = \langle f(t), g(t)\rangle = f(t)\mathbf{i} + g(t)\mathbf{j}$$

and

$$\mathbf{r}(t) = \langle f(t), g(t), h(t)\rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k},$$

are **vector-valued functions** or simply **vector functions**. As shown in **FIGURE 9.1.1**, for a given value of the parameter, say t_0, the vector $\mathbf{r}(t_0)$ is the position vector of a point P on a curve C. In other words, as the parameter t varies, we can envision the curve C being traced out by the moving arrowhead of $\mathbf{r}(t)$.

We have already seen an example of parametric equations, as well as the vector function of a space curve, in Section 7.5, when we discussed the line in 3-space.

EXAMPLE 1 Circular Helix

Graph the curve traced by the vector function

$$\mathbf{r}(t) = 2\cos t\,\mathbf{i} + 2\sin t\,\mathbf{j} + t\mathbf{k}, \quad t \ge 0.$$

SOLUTION The parametric equations of the curve are $x = 2\cos t$, $y = 2\sin t$, $z = t$. By eliminating the parameter t from the first two equations:

$$x^2 + y^2 = (2\cos t)^2 + (2\sin t)^2 = 2^2$$

we see that a point on the curve lies on the circular cylinder $x^2 + y^2 = 4$. As seen in **FIGURE 9.1.2** and the accompanying table, as the value of t increases, the curve winds upward in a spiral or **circular helix**.

t	x	y	z
0	2	0	0
$\pi/2$	0	2	$\pi/2$
π	-2	0	π
$3\pi/2$	0	-2	$3\pi/2$
2π	2	0	2π
$5\pi/2$	0	2	$5\pi/2$
3π	-2	0	3π
$7\pi/2$	0	-2	$7\pi/2$
4π	2	0	4π
$9\pi/2$	0	2	$9\pi/2$

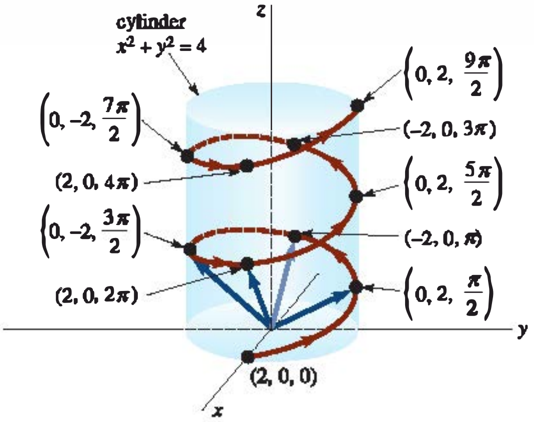

FIGURE 9.1.2 Circular helix in Example 1 ≡

The curve in Example 1 is a special case of the vector function

$$\mathbf{r}(t) = a\cos t\,\mathbf{i} + b\sin t\,\mathbf{j} + ct\,\mathbf{k}, \quad a > 0, \quad b > 0, \quad c > 0,$$

which describes an **elliptical helix**. When $a = b$, the helix is circular. The **pitch** of a helix is defined to be the number $2\pi c$. Problems 9 and 10 in Exercises 9.1 illustrate two other kinds of helixes.

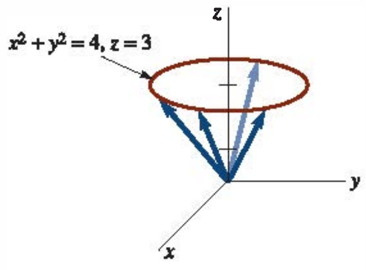

FIGURE 9.1.3 Curve in Example 2

EXAMPLE 2 Circle in a Plane

Graph the curve traced by the vector function

$$\mathbf{r}(t) = 2\cos t\,\mathbf{i} + 2\sin t\,\mathbf{j} + 3\mathbf{k}.$$

SOLUTION The parametric equations of this curve are $x = 2\cos t$, $y = 2\sin t$, $z = 3$. As in Example 1, we see that a point on the curve must also lie on the cylinder $x^2 + y^2 = 4$. However, since the z-coordinate of any point has the constant value $z = 3$, the vector function $\mathbf{r}(t)$ traces out a circle 3 units above the xy-plane. See **FIGURE 9.1.3**. ≡

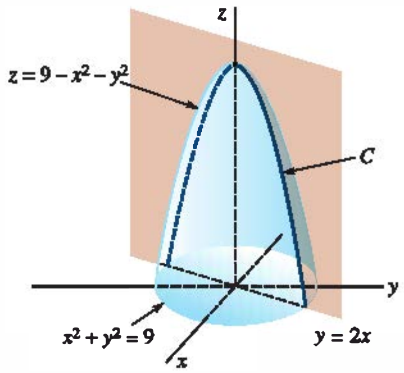

FIGURE 9.1.4 Curve in Example 3

EXAMPLE 3 Curve of Intersection

Find the vector function that describes the curve C of intersection of the plane $y = 2x$ and the paraboloid $z = 9 - x^2 - y^2$.

SOLUTION We first parameterize the curve C of intersection by letting $x = t$. It follows that $y = 2t$ and $z = 9 - t^2 - (2t)^2 = 9 - 5t^2$. From the parametric equations $x = t$, $y = 2t$, $z = 9 - 5t^2$, we see that a vector function describing the trace of the paraboloid in the plane $y = 2x$ is given by $\mathbf{r}(t) = t\mathbf{i} + 2t\mathbf{j} + (9 - 5t^2)\mathbf{k}$. See **FIGURE 9.1.4**. ≡

Limits, Continuity, and Derivatives

The fundamental notion of the **limit** of a vector function $\mathbf{r}(t) = \langle f(t), g(t), h(t)\rangle$ is defined in terms of the limits of the component functions.

Definition 9.1.1 Limit of a Vector Function

If $\lim_{t\to a} f(t)$, $\lim_{t\to a} g(t)$, and $\lim_{t\to a} h(t)$ exist, then

$$\lim_{t\to a}\mathbf{r}(t) = \left\langle \lim_{t\to a} f(t), \lim_{t\to a} g(t), \lim_{t\to a} h(t)\right\rangle.$$

The notation $t \to a$ in Definition 9.1.1 can, of course, be replaced by $t \to a^+$, $t \to a^-$, $t \to \infty$, or $t \to -\infty$.

As an immediate consequence of Definition 9.1.1, we have the following result.

Theorem 9.1.1 Properties of Limits

If $\lim_{t\to a}\mathbf{r}_1(t) = \mathbf{L}_1$ and $\lim_{t\to a}\mathbf{r}_2(t) = \mathbf{L}_2$, then

(*i*) $\lim_{t\to a} c\mathbf{r}_1(t) = c\mathbf{L}_1$, c a scalar

(*ii*) $\lim_{t\to a} [\mathbf{r}_1(t) + \mathbf{r}_2(t)] = \mathbf{L}_1 + \mathbf{L}_2$

(*iii*) $\lim_{t\to a} \mathbf{r}_1(t)\cdot\mathbf{r}_2(t) = \mathbf{L}_1\cdot\mathbf{L}_2$.

Definition 9.1.2 Continuity of a Vector Function

A vector function $\mathbf{r}$ is said to be **continuous** at $t = a$ if

(*i*) $\mathbf{r}(a)$ is defined, (*ii*) $\lim_{t\to a}\mathbf{r}(t)$ exists, and (*iii*) $\lim_{t\to a}\mathbf{r}(t) = \mathbf{r}(a)$.

Equivalently, $\mathbf{r}(t)$ is continuous at $t = a$ if and only if the component functions f, g, and h are continuous there.

Definition 9.1.3 Derivative of Vector Function

The **derivative** of a vector function $\mathbf{r}$ is

$$\mathbf{r}'(t) = \lim_{\Delta t \to 0} \frac{1}{\Delta t}[\mathbf{r}(t + \Delta t) - \mathbf{r}(t)] \tag{2}$$

for all t for which the limit exists.

The derivative of $\mathbf{r}$ is also written $d\mathbf{r}/dt$. The next theorem will show that on a practical level the derivative of a vector function is obtained by simply differentiating its component functions.

Theorem 9.1.2 Differentiation of Components

If $\mathbf{r}(t) = \langle f(t), g(t), h(t)\rangle$, where f, g, and h are differentiable, then

$$\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t)\rangle.$$

PROOF: From (2) we have

$$\begin{aligned}\mathbf{r}'(t) &= \lim_{\Delta t \to 0} \frac{1}{\Delta t}\Big[\langle f(t + \Delta t), g(t + \Delta t), h(t + \Delta t)\rangle - \langle f(t), g(t), h(t)\rangle\Big] \\ &= \lim_{\Delta t \to 0}\left\langle \frac{f(t + \Delta t) - f(t)}{\Delta t}, \frac{g(t + \Delta t) - g(t)}{\Delta t}, \frac{h(t + \Delta t) - h(t)}{\Delta t}\right\rangle.\end{aligned}$$

Taking the limit of each component yields the desired result. ≡

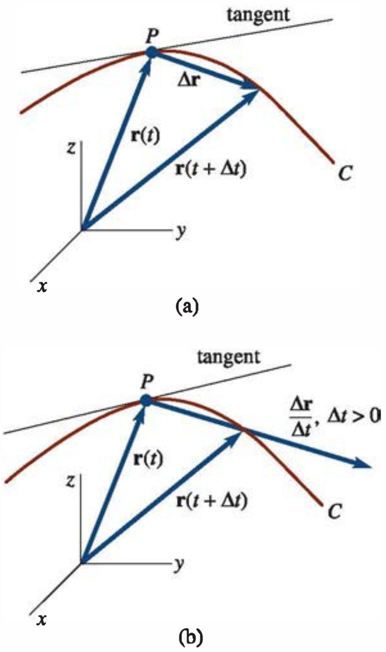

FIGURE 9.1.5 Vector $\mathbf{r}'(t)$ is tangent to curve C at P

Smooth Curves

When the component functions of a vector function $\mathbf{r}$ have continuous first derivatives and $\mathbf{r}'(t) \neq \mathbf{0}$ for all t in the open interval (a, b), then $\mathbf{r}$ is said to be a **smooth function** and the curve C traced by $\mathbf{r}$ is called a **smooth curve**.

Geometric Interpretation of $\mathbf{r}'(t)$

If the vector $\mathbf{r}'(t)$ is not $\mathbf{0}$ at a point P, then it may be drawn *tangent to the curve* at P. As seen in **FIGURE 9.1.5**, the vectors

$$\Delta\mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t) \quad \text{and} \quad \frac{\Delta\mathbf{r}}{\Delta t} = \frac{1}{\Delta t}[\mathbf{r}(t + \Delta t) - \mathbf{r}(t)]$$

are parallel. If we assume $\lim_{\Delta t \to 0} \Delta\mathbf{r}/\Delta t$ exists, it seems reasonable to conclude that as $\Delta t \to 0$, $\mathbf{r}(t)$ and $\mathbf{r}(t + \Delta t)$ become close and, as a consequence, the limiting position of the vector $\Delta\mathbf{r}/\Delta t$ is the tangent line at P. Indeed, the tangent line at P is *defined* as that line through P parallel to $\mathbf{r}'(t)$.

EXAMPLE 4 Tangent Vectors

Graph the curve C that is traced by a point P whose position is given by $\mathbf{r}(t) = \cos 2t\mathbf{i} + \sin t\mathbf{j}$, $0 \le t \le 2\pi$. Graph $\mathbf{r}'(0)$ and $\mathbf{r}'(\pi/6)$.

SOLUTION By clearing the parameter from the parametric equations $x = \cos 2t$, $y = \sin t$, $0 \le t \le 2\pi$, we find that C is the parabola $x = 1 - 2y^2$, $-1 \le x \le 1$. From $\mathbf{r}'(t) = -2\sin 2t\mathbf{i} + \cos t\mathbf{j}$ we find

$$\mathbf{r}'(0) = \mathbf{j} \quad \text{and} \quad \mathbf{r}'\left(\frac{\pi}{6}\right) = -\sqrt{3}\,\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j}.$$

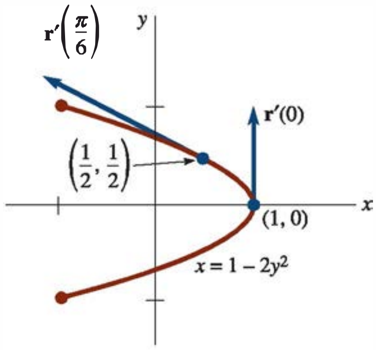

FIGURE 9.1.6 Tangent vectors in Example 4

In **FIGURE 9.1.6** these vectors are drawn tangent to the curve C at $(1, 0)$ and $(\frac{1}{2}, \frac{1}{2})$, respectively. ≡

EXAMPLE 5 Tangent Line

Find parametric equations of the tangent line to the graph of the curve C whose parametric equations are $x = t^2, y = t^2 - t, z = -7t$ at $t = 3$.

SOLUTION The vector function that gives the position of a point P on the curve is given by $\mathbf{r}(t) = t^2\mathbf{i} + (t^2 - t)\mathbf{j} - 7t\mathbf{k}$. Now,

$$\mathbf{r}'(t) = 2t\mathbf{i} + (2t - 1)\mathbf{j} - 7\mathbf{k} \quad \text{and so} \quad \mathbf{r}'(3) = 6\mathbf{i} + 5\mathbf{j} - 7\mathbf{k},$$

which is tangent to C at the point whose position vector is

$$\mathbf{r}(3) = 9\mathbf{i} + 6\mathbf{j} - 21\mathbf{k};$$

that is, $P(9, 6, -21)$. Using the components of $\mathbf{r}'(3)$, we see that parametric equations of the tangent line are $x = 9 + 6t, y = 6 + 5t, z = -21 - 7t$. ≡

Higher-Order Derivatives Higher-order derivatives of a vector function are also obtained by differentiating its components. In the case of the **second derivative**, we have

$$\mathbf{r}''(t) = \langle f''(t), g''(t), h''(t)\rangle = f''(t)\mathbf{i} + g''(t)\mathbf{j} + h''(t)\mathbf{k}.$$

EXAMPLE 6 Derivative of a Vector Function

If $\mathbf{r}(t) = (t^3 - 2t^2)\mathbf{i} + 4t\mathbf{j} + e^{-t}\mathbf{k}$, then

$$\mathbf{r}'(t) = (3t^2 - 4t)\mathbf{i} + 4\mathbf{j} - e^{-t}\mathbf{k} \quad \text{and} \quad \mathbf{r}''(t) = (6t - 4)\mathbf{i} + e^{-t}\mathbf{k}.$$ ≡

Theorem 9.1.3 Chain Rule

If $\mathbf{r}$ is a differentiable vector function and $s = u(t)$ is a differentiable scalar function, then the derivative of $\mathbf{r}(s)$ with respect to t is

$$\frac{d\mathbf{r}}{dt} = \frac{d\mathbf{r}}{ds}\frac{ds}{dt} = \mathbf{r}'(s)u'(t).$$

EXAMPLE 7 Chain Rule

If $\mathbf{r}(s) = \cos 2s\,\mathbf{i} + \sin 2s\,\mathbf{j} + e^{-3s}\mathbf{k}$, where $s = t^4$, then

$$\begin{aligned}\frac{d\mathbf{r}}{dt} &= [-2\sin 2s\,\mathbf{i} + 2\cos 2s\,\mathbf{j} - 3e^{-3s}\mathbf{k}]4t^3\\ &= -8t^3\sin(2t^4)\mathbf{i} + 8t^3\cos(2t^4)\mathbf{j} - 12t^3e^{-3t^4}\mathbf{k}.\end{aligned}$$ ≡

Details of the proof of the next theorem are left as exercises.

Theorem 9.1.4 Rules of Differentiation

Let $\mathbf{r}_1$ and $\mathbf{r}_2$ be differentiable vector functions and $u(t)$ a differentiable scalar function.

(*i*) $\dfrac{d}{dt}[\mathbf{r}_1(t) + \mathbf{r}_2(t)] = \mathbf{r}_1'(t) + \mathbf{r}_2'(t)$

(*ii*) $\dfrac{d}{dt}[u(t)\mathbf{r}_1(t)] = u(t)\mathbf{r}_1'(t) + u'(t)\mathbf{r}_1(t)$

(*iii*) $\dfrac{d}{dt}[\mathbf{r}_1(t)\cdot\mathbf{r}_2(t)] = \mathbf{r}_1(t)\cdot\mathbf{r}_2'(t) + \mathbf{r}_1'(t)\cdot\mathbf{r}_2(t)$

(*iv*) $\dfrac{d}{dt}[\mathbf{r}_1(t)\times\mathbf{r}_2(t)] = \mathbf{r}_1(t)\times\mathbf{r}_2'(t) + \mathbf{r}_1'(t)\times\mathbf{r}_2(t)$.

Since the cross product of two vectors is not commutative, the order in which $\mathbf{r}_1$ and $\mathbf{r}_2$ appear in part (*iv*) of Theorem 9.1.4 must be strictly observed. Note of caution.

Integrals of Vector Functions

If f, g, and h are integrable, then the indefinite and definite integrals of a vector function $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ are defined, respectively, by

$$\int \mathbf{r}(t)\,dt = \left[\int f(t)\,dt\right]\mathbf{i} + \left[\int g(t)\,dt\right]\mathbf{j} + \left[\int h(t)\,dt\right]\mathbf{k}$$

$$\int_a^b \mathbf{r}(t)\,dt = \left[\int_a^b f(t)\,dt\right]\mathbf{i} + \left[\int_a^b g(t)\,dt\right]\mathbf{j} + \left[\int_a^b h(t)\,dt\right]\mathbf{k}.$$

The indefinite integral of $\mathbf{r}$ is another vector function $\mathbf{R} + \mathbf{c}$ such that $\mathbf{R}'(t) = \mathbf{r}(t)$.

EXAMPLE 8 Integral of a Vector Function

If $\mathbf{r}(t) = 6t^2\mathbf{i} + 4e^{-2t}\mathbf{j} + 8\cos 4t\mathbf{k}$,

then
$$\int \mathbf{r}(t)\,dt = \left[\int 6t^2\,dt\right]\mathbf{i} + \left[\int 4e^{-2t}\,dt\right]\mathbf{j} + \left[\int 8\cos 4t\,dt\right]\mathbf{k}$$

$$= [2t^3 + c_1]\mathbf{i} + [-2e^{-2t} + c_2]\,\mathbf{j} + [2\sin 4t + c_3]\mathbf{k}$$

$$= 2t^3\mathbf{i} - 2e^{-2t}\mathbf{j} + 2\sin 4t\mathbf{k} + \mathbf{c},$$

where $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$. ≡

Length of a Space Curve

If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ is a smooth function, then it can be shown that the **length** of the smooth curve traced by $\mathbf{r}$ is given by

$$s = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2}\,dt = \int_a^b \|\mathbf{r}'(t)\|\,dt. \tag{3}$$

Arc Length as a Parameter

A curve in the plane or in space can be parameterized in terms of the arc length s.

EXAMPLE 9 Example 1 Revisited

Consider the helix of Example 1. Since $\|\mathbf{r}'(t)\| = \sqrt{5}$, it follows from (3) that the length of the curve from $\mathbf{r}(0)$ to an arbitrary point $\mathbf{r}(t)$ is

$$s = \int_0^t \sqrt{5}\,du = \sqrt{5}t,$$

where we have used u as a dummy variable of integration. Using $t = s/\sqrt{5}$, we obtain a vector equation of the helix as a function of arc length:

$$\mathbf{r}(s) = 2\cos\frac{s}{\sqrt{5}}\mathbf{i} + 2\sin\frac{s}{\sqrt{5}}\mathbf{j} + \frac{s}{\sqrt{5}}\mathbf{k}. \tag{4}$$

Parametric equations of the helix are then

$$f(s) = 2\cos\frac{s}{\sqrt{5}}, \quad g(s) = 2\sin\frac{s}{\sqrt{5}}, \quad h(s) = \frac{s}{\sqrt{5}}.$$ ≡

The derivative of a vector function $\mathbf{r}(t)$ with respect to the parameter t is a tangent vector to the curve traced by $\mathbf{r}$. However, if the curve is parameterized in terms of arc length s, then $\mathbf{r}'(s)$ *is a unit tangent vector*. To see this, let a curve be described by $\mathbf{r}(s)$, where s is arc length. From (3), the length of the curve from $\mathbf{r}(0)$ to $\mathbf{r}(s)$ is $s = \int_0^s \|\mathbf{r}'(u)\|\,du$. Differentiation of this last equation with respect to s then yields $\|\mathbf{r}'(s)\| = 1$.